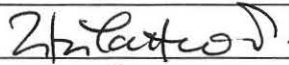
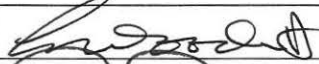
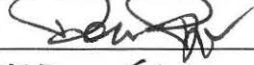


		CONTROLLED FORM		
TITLE: GXP and Non-GXP Study Report Review and Approval Form				
DOCUMENT NO. GE003FM01	VERSION 02	ISSUE DATE 17 Oct 2014	EFFECTIVE DATE 20 Oct 2014	PAGE 2 of 2

☐ Vendor Study Protocol		☐ Vendor Study Report		☒ Lpath Study Report	
Lpath Study No.: LT3114-PHA-015-R		Version No.: 01		☒ Non GXP GCP	☐ GLP GMP
Product Name: LT3114		Test Article: B3			
Study Title: Pharamcokinetic evaluation of B3 in C57Bl/6 mice					
Study Vendor/Site: ☐ N/A LPath		Study Director: ☐ N/A Rosalia Matteo		Study Contributor(s): ☐ N/A Kelli Moreno	
Vendor Study No./Version: ☒ N/A		Study Dates: Nov-Dec 2009		Report Date: May 2013	
Lab Notebook Reference: LNB#189 pp.177-180					☐ N/A
Executive summary of study results: ☐ See Attached ☐ N/A The pharmacokinetics of the murine anti-LPA antibody, B3, was examined in C57Bl/6 mice. Following intravenous administration the AUC was 129746.15 hr*mg/L with a Cmax of 670.92 mg/L and a terminal half-life of 9.45 days. Following subcutaneous administration the AUC was 104511.90 hr*mg/L with a Cmax of 417.2 mg/L with a terminal half-life of approximately 7.0 days. The bioavailability was 80.6%.					

LPATH APPROVAL			
Function	Name (Print)	Signature	Date
Study Director/Monitor	Rosalia Matteo		11/14/2014
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Pharmacokinetic evaluation of B3 in C57Bl/6 mice

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Study Monitor	Rosalia Matteo
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1 SUMMARY

The pharmacokinetics of the murine anti-LPA antibody, B3, was examined in C57Bl/6 mice. Following intravenous administration the AUC was 129746.15 hr*mg/L with a Cmax of 670.92 mg/L and a terminal half-life of 9.45 days. Following subcutaneous administration the AUC was 104511.90 hr*mg/L with a Cmax of 417.2 mg/L with a terminal half-life of approximately 7.0 days. The bioavailability was 80.6%.

2 INTRODUCTION

The overall goal was to examine the pharmacokinetics (PK) of the murine anti-LPA antibody (B3) in C57Bl/6 mice. Systemic concentrations of the B3 antibody were measured in these mice by either an intravenous (iv) or subcutaneous (sc) route of administration and use the measured concentrations to calculate several PK parameters useful in further development of the antibody.

3 MATERIALS AND METHODS

3.1 Test Article and Control

3.2 Study Design

In this pharmacokinetic study, mice were injected with either an SC or an IV bolus dose of the murine B3 mAb at a dose of 30 mg/kg. Mice (3 per time point) were sacrificed at designated time points (from 0 hours to 21 Days) and blood samples were collected by terminal cardiac puncture. The relative concentrations of B3 at these times were determined using a direct-binding ELISA.

The dose and volume for each animal was based on the most recent body weight taken the day prior to dosing. The animals were temporarily restrained but not sedated for dosing.

Species:	Mus musculus
Strain:	C57Bl/6
Sex:	Female
Weight:	18.0 – 22.0 g
Age:	6 - 8 weeks approximately
Number:	60
Acclimation:	7 days

3.3 Blood Collection and Analysis

At each time point, approximately 1 mL of whole blood was collected via terminal cardiac puncture into lithium heparin tubes and processed for collection of plasma by centrifuging at approximately 5000 rpm for approximately 10 minutes at 4°C. Plasma samples were immediately frozen in dry ice and stored at -70±10°C until assay date.

Blood was sampled at the following times:

- IV group: 0, 0.5, 3, 6, 12, 24 hours; 3, 5, 7, 14, and 21 Days post-dose
- SC group: 0, 0.5, 3, 6, 12, 24 hours; 3, 5, 7, 14, and 21 Days post-dose

3.4 Direct Binding ELISA

A 96 well plate was coated with 100 µl of 1 µg/ml of (C12)LPA-BSA-SMCC for 1 hour at 37 degrees. The plates were blocked with 150 µl of PBS-BSA-tween for 1 hour at room temperature. 100 µl of standard antibody was added starting at 4 µg/ml and diluting 2-fold across the plate. And 100 µl of plasma sample was added at 1:100, 1:1000, 1:2000, 1:4000, 1:8000, and 1:16000 dilutions. This step was incubated at room temperature for 1 hour. Next, 100 µl of a peroxidase conjugated anti-mouse IgG (H+L) (Jackson ImmunoResearch 115-035-062) was added at a 1:40K dilution and incubated for 1 hour at room temperature. Finally, the plates were developed with 100 µl cold TMB substrate for 12 minutes at room temperature. And the reaction was stopped with 100 µl of 1M H₂SO₄. The plates were then read at 450 nm.

3.5 Data Analysis

The relative concentrations of B3 were determined by direct-binding ELISA and data was analyzed using WinNonlin.

4 RESULTS

Antibody concentrations measured in rat plasma following intravenous or subcutaneous dosing are shown in Figure 1 and Figure 2 respectively. Lines of best fit from WinNonlin analysis are shown in Figure 3 and Figure 4 with the associated calculated pharmacokinetic parameters (Table 1 and Table 2). See also Appendix 1 and 2 for full WinNonlin analyses.

Figure 1 Plasma Concentrations of B3 following intravenous dosing

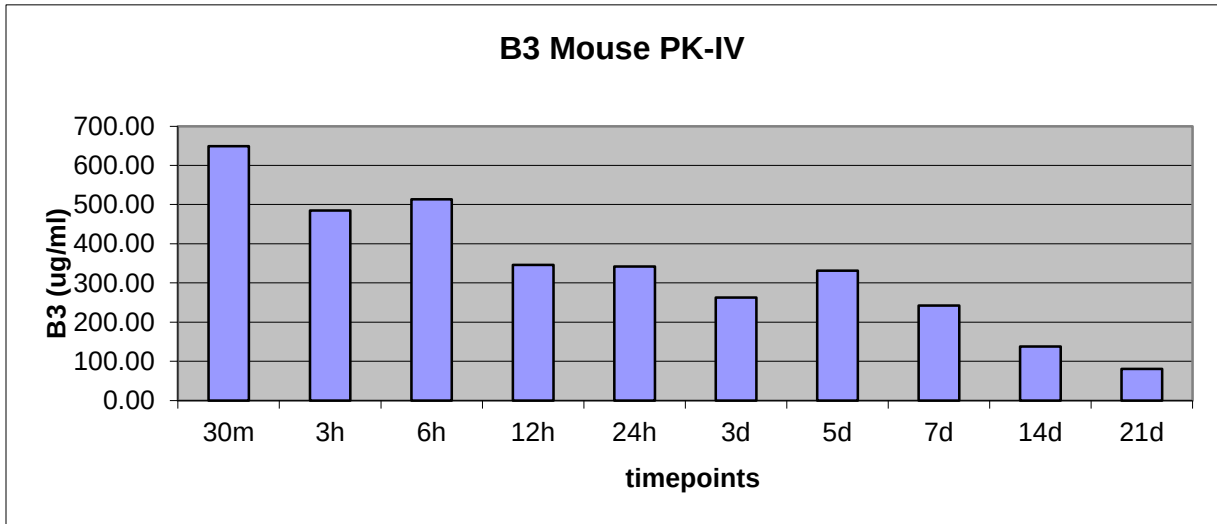


Figure 2 Plasma Concentrations of B3 following subcutaneous dosing

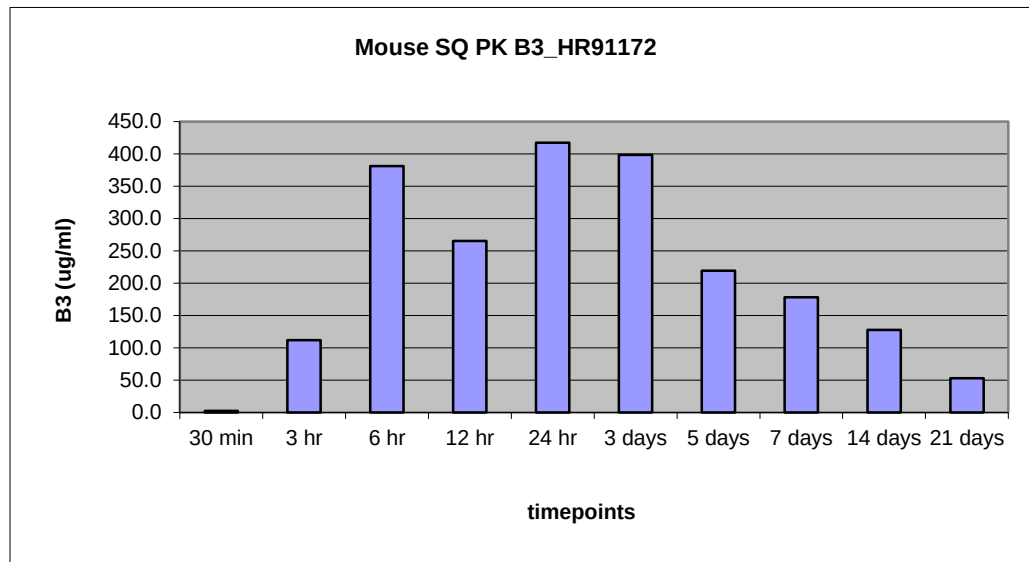


Figure 3 Lines of best fit following WinNonlin analysis (intravenous)

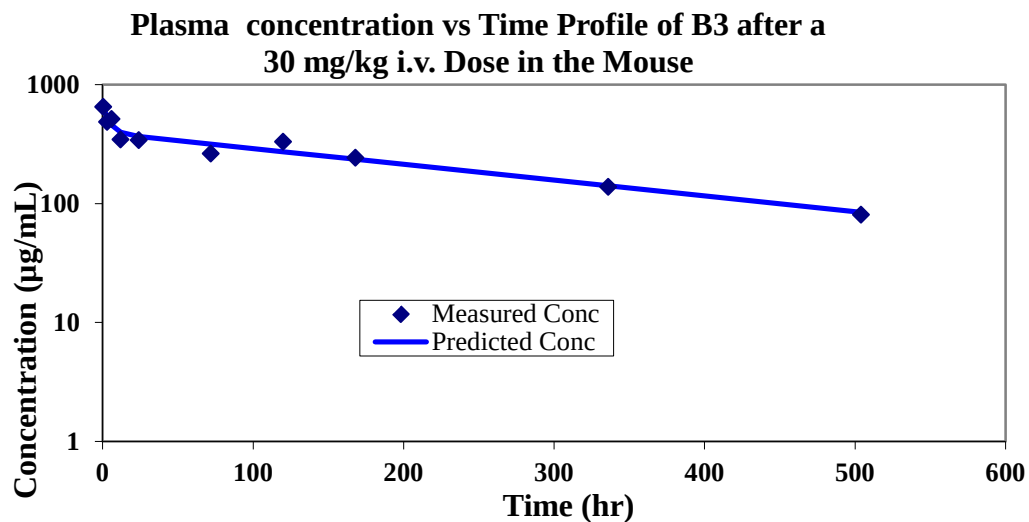
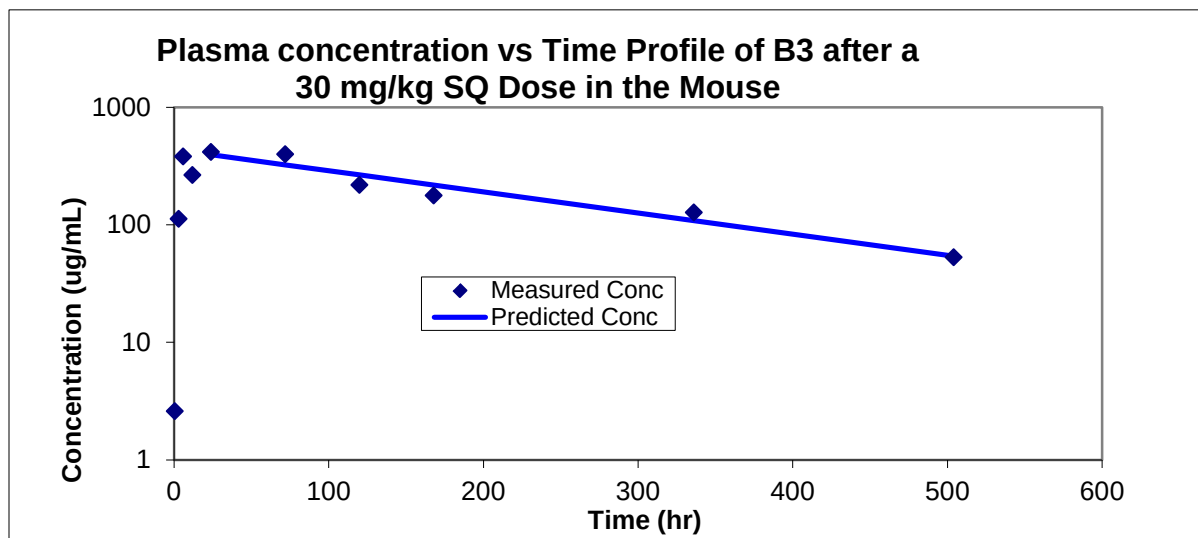


Table 1 Pharmacokinetic parameters following intravenous dosing

Parameter	Units	Estimate	Standard Error	%CV
AUC	hr*mg/L	129746.15	9304.26	7.17
K10_HL	hr	134.04	23.85	17.79
Alpha	1/hr	0.23	0.16	72.73
Beta	1/hr	0.00306	0.00035	11.36
Alpha_HL	hr	3.07	2.23	72.69
Beta_HL	hr	226.72	25.73	11.35
Beta_HL	Days	9.45	1.07	11.35
A	mg/L	278.02	109.66	39.44
B	mg/L	392.90	35.69	9.08
Cmax	mg/L	670.92	109.40	16.31
CL	L/hr/kg	0.000231	0.000017	7.18
AUMC	hr*hr*mg/L	42041204.00	7034716.59	16.73
MRT	hr	324.03	35.95	11.09
Vss	L/kg	0.0749	0.0062	8.3
V2	L/kg	0.0302	0.0089	29.33
CLD2	L/hr/kg	0.0040	0.0030	74.18

Figure 4 Lines of best fit following WinNonlin analysis (subcutaneous)**Table 2** Pharmacokinetic parameters following subcutaneous dosing

Parameter	Units	Estimate
AUC	hr*mg/L	104511.90
Beta_HL	hr	167.6
Beta_HL	Days	7.0
Cmax	mg/L	417.2
CL/F	L/hr/kg	0.0003
Vz/F	L/kg	0.0692
F	% bioavailable	80.6

5 CONCLUSION

The pharmacokinetics of the murine anti-LPA antibody, B3, was examined in C57Bl/6 mice. Following intravenous administration the AUC was 129746.15 hr*mg/L with a Cmax of 670.92 mg/L and a terminal half-life of 9.45 days. Following subcutaneous administration the AUC was 104511.90 hr*mg/L with a Cmax of 417.2 mg/L and a terminal half-life of approximately 7.0 days. The bioavailability was 80.6%.

6 DATA ARCHIVE

LPath Notebook #189 pp.177-180

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2010\LPA\PK\B3 PK rats&mice\Final QCd data and PK reports

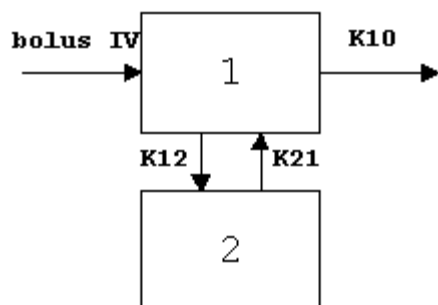
7 APPENDICES

Appendix 1

30mg/kg intravenous injection of B# in C57Bl/6 mice WinNonlin Compartmental Modeling Analysis

WinNonlin compiled model:

2 compartment IV-Bolus, micro-constants, no lag time, 1st order elimination



$$C(T) = A \cdot \exp(-\alpha \cdot T) + B \cdot \exp(-\beta \cdot T)$$

Settings for analysis:

Input Workbook: B3 Mouse PK

Input Worksheet: 30 mg/kg IV

Input Sort Keys: [none]

Nelder-Mead method used

Iterative reweighting used during minimization process

Convergence criteria of 0.0001 used during minimization process

500 maximum iterations allowed during minimization process

Input data:

Time (hr)	Conc (mg/L)
0.5	648.7
3	484.7
6	513.46
12	345.97
24	341.62
72	262.34
120	331.47
168	242.26
336	138.04
504	80.72

Output data:

Initial Parameters

Parameter	Value	Lower	Upper
V1	4.22E-02	0	0.4217
K10	5.55E-03	0	5.55E-02
K12	0.1786	0	1.786
K21	0.2418	0	2.418

Final Parameters

V1	V1 Std	V1 CV	V1 Univ	V1 Univ	V1 Plan	V1 Plan	K10	K10 Std	K10 CV	K10 Un	K10 Un	K10 Pla	K10 Pla	K12	K12 Std	K12 CV	K12 Un	K12 Un	K12 Pla	K12 Pla	K21	K21 Std	K21 CV	K21 Un	K21 Un	K21 Pla	K21 Pla
0.044715	0.0072	0.163	0.002	0.006	0.001	0.007	0.000	0.000	0.178	0.000	0.000	0.000	0.000	0.000	0.000	0.086	-	0.028	-	0.042	0.013	0.009	0.068	-	0.035	-	0.052
			0.008	0.025	0.030	0.063	0.051	0.009	0.1	0.029	0.074	0.111	0.091	0.000	0.081	0.3	0.10	0.12	0.24	0.084	0.033	0.007	0.08	0.008	0.054	0.025	0.065
			0.057	0.073	0.096	0.33	0.071	0.021		0.17	0.25	0.81	0.61	0.071	0.16		0.10	0.13	0.83	0.94	0.30	0.071		0.087	0.039	0.099	0.80
																	0.71		0.52					0.078		0.19	

Non-Transposed Final Parameters

Parameter	Units	Estimate	StdError	CV%	UnivarCI _Lower	UnivarCI _Upper	PlanarCI _Lower	PlanarCI _Upper
V1	L/kg	0.044715	0.007298	16.32	0.026857	0.062573	0.013096	0.076333
K10	1/hr	0.005171	0.000921	17.81	0.002917	0.007425	0.001181	0.009161
K12	1/hr	0.090071	0.078116	86.73	-0.101071	0.281213	-0.248352	0.428494
K21	1/hr	0.133330	0.090771	68.08	-0.088778	0.355439	-0.259919	0.526580

Dosing

Constant	Value
Number of Doses	1
Dose #1	30
Time of Dose #1	0

Correlation Matrix

Parameter	V1	K10	K12	K21
V1	1			
K10	-0.915203	1		
K12	-0.807261	0.748177	1	
K21	-0.478882	0.453221	0.875752	1

Eigenvalues

Number	Value
1	1.50E+05
2	3111
3	22.76
4	1.51

Variance-Covariance Matrix

Parameter	V1	K10	K12	K21
V1	5.33E-05			
K10	-6.15E-06	8.48E-07		
K12	-4.60E-04	5.38E-05	6.10E-03	
K21	-3.17E-04	3.79E-05	6.21E-03	8.24E-03

Summary Table

Time_obs (hr)	Conc_obs (mg/L)	Time (hr)	Conc (mg/L)	Predicted (mg/L)	Residual (mg/L)	Weight	SE_Yhat	Standard_Res
0.5	648.7	0.5000	648.7000	640.6732	8.0268	0.0000	86.1854	0.2622
3	484.7	3.0000	484.7000	530.6479	-45.9479	0.0000	47.4379	-0.7779
6	513.46	6.0000	513.4600	457.6071	55.8529	0.0000	47.1634	1.2353
12	345.97	12.0000	345.9700	397.3126	-51.3426	0.0000	30.2183	-1.0692
24	341.62	24.0000	341.6200	366.3402	-24.7202	0.0000	28.5676	-0.5640
72	262.34	72.0000	262.3400	315.2682	-52.9282	0.0000	23.0999	-1.3697
120	331.47	120.0000	331.4700	272.2378	59.2322	0.0000	17.5338	1.7072
168	242.26	168.0000	242.2600	235.0806	7.1794	0.0000	13.8753	0.2349
336	138.04	336.0000	138.0400	140.6553	-2.6153	0.0001	10.4683	-0.1526
504	80.72	504.0000	80.7200	84.1580	-3.4380	0.0001	10.1037	-0.5293

Diagnostics

Function	Item	Value
1	CSS	270020
1	WCSS	3.55036
1	SSR	14922.4
1	WSSR	0.122213
1	S	0.142719
1	DF	6
1	CORR_(OBS, PRED)	0.9727
1	WT_CORR_(OBS, PRED)	0.9829
1	AIC	-13.01991
	SBC	-11.80957

Partial Derivatives

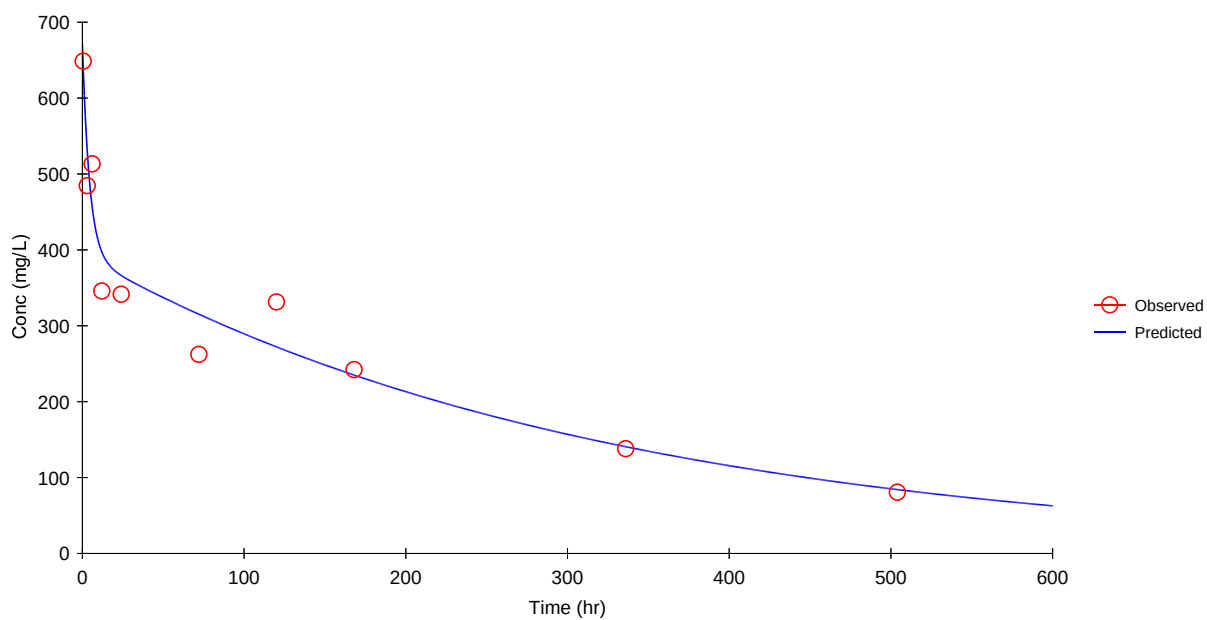
Function	Time (hr)	V1	K10	K12	K21
1	0.5	-22.36399255	-0.49975312	-0.48321815	0.01092888
1	3	-22.36399255	-2.95098604	-2.39236591	0.32841372
1	6	-22.36399255	-5.66633706	-3.68909140	1.00182058
1	12	-22.36399255	-10.23690069	-4.41583035	2.16753007
1	24	-22.36399255	-17.73655074	-4.23003663	2.85684755
1	72	-22.36399255	-45.89444148	-3.52379916	2.50671435
1	120	-22.36399255	-74.01206726	-2.86421149	2.05103714
1	168	-22.36399255	-102.13377860	-2.20466792	1.59532615
1	336	-22.36399255	-200.59196400	0.10342612	0.00011954
1	504	-22.36399255	-299.10025080	2.41104041	-1.59542643

Secondary Parameters

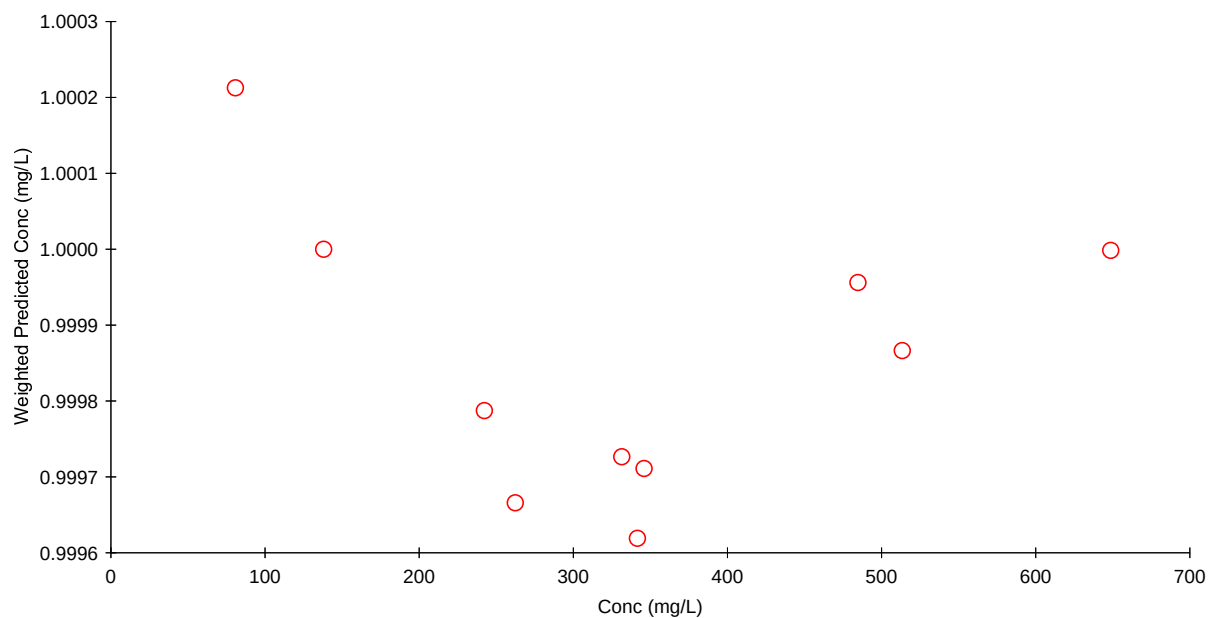
Non-Transposed Secondary Parameters

Parameter	Units	Estimate	StdError	CV%
AUC	hr*mg/L	129746.146999	9304.258819	7.17
K10_HL	hr	134.044604	23.852686	17.79
Alpha	1/hr	0.225515	0.164016	72.73
Beta	1/hr	0.003057	0.000347	11.36
Alpha_HL	hr	3.073615	2.234303	72.69
Beta_HL	hr	226.723232	25.728837	11.35
A	mg/L	278.023835	109.656305	39.44
B	mg/L	392.895941	35.692835	9.08
Cmax	mg/L	670.919777	109.396243	16.31
CL	L/hr/kg	0.000231	0.000017	7.18
AUMC	hr*hr*mg/L	42041203.999007	7034716.593301	16.73
MRT	hr	324.026609	35.948360	11.09
Vss	L/kg	0.074922	0.006215	8.30
V2	L/kg	0.030207	0.008859	29.33
CLD2	L/hr/kg	0.004028	0.002988	74.18

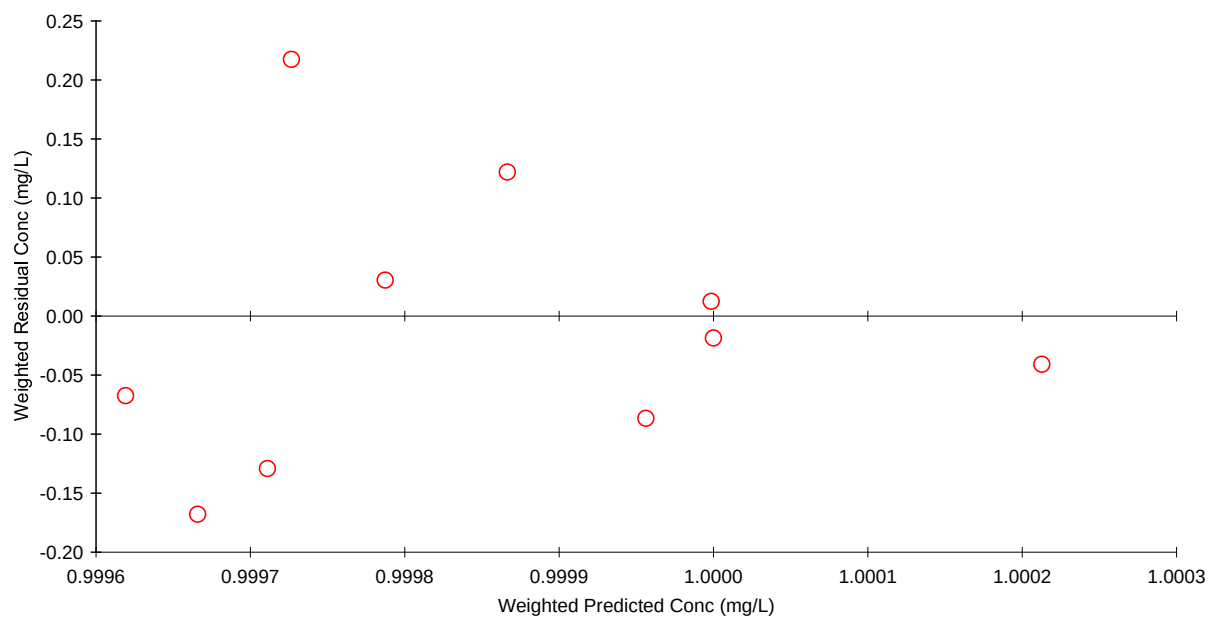
X vs. Observed Y and Predicted Y



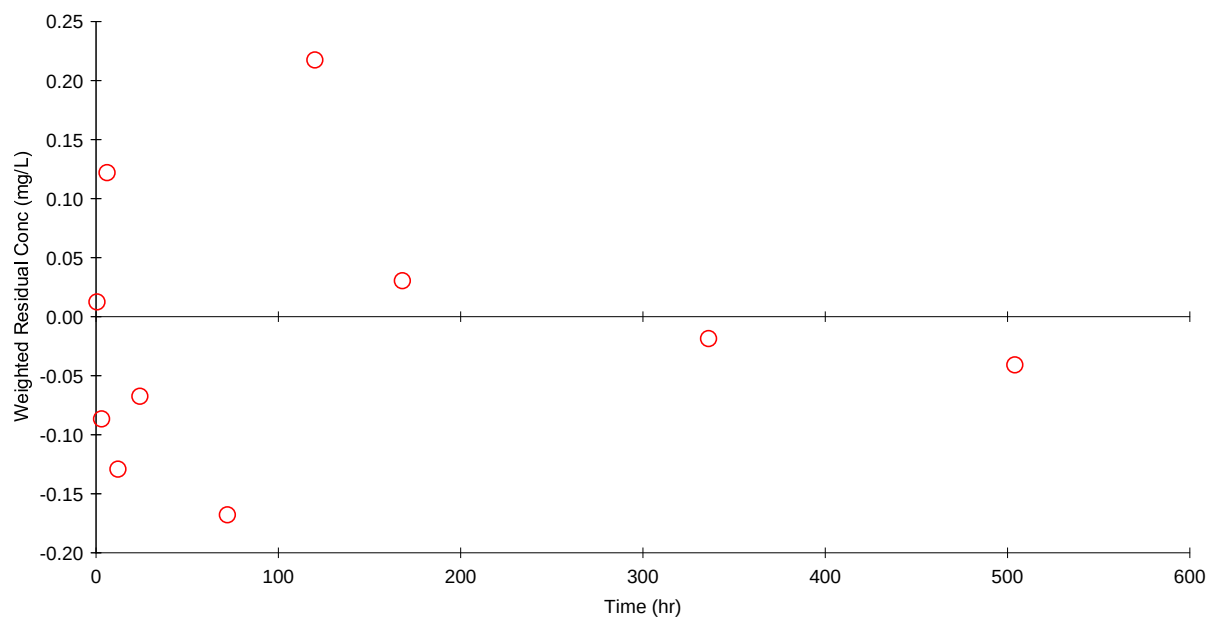
Observed Y vs. Weighted Predicted Y



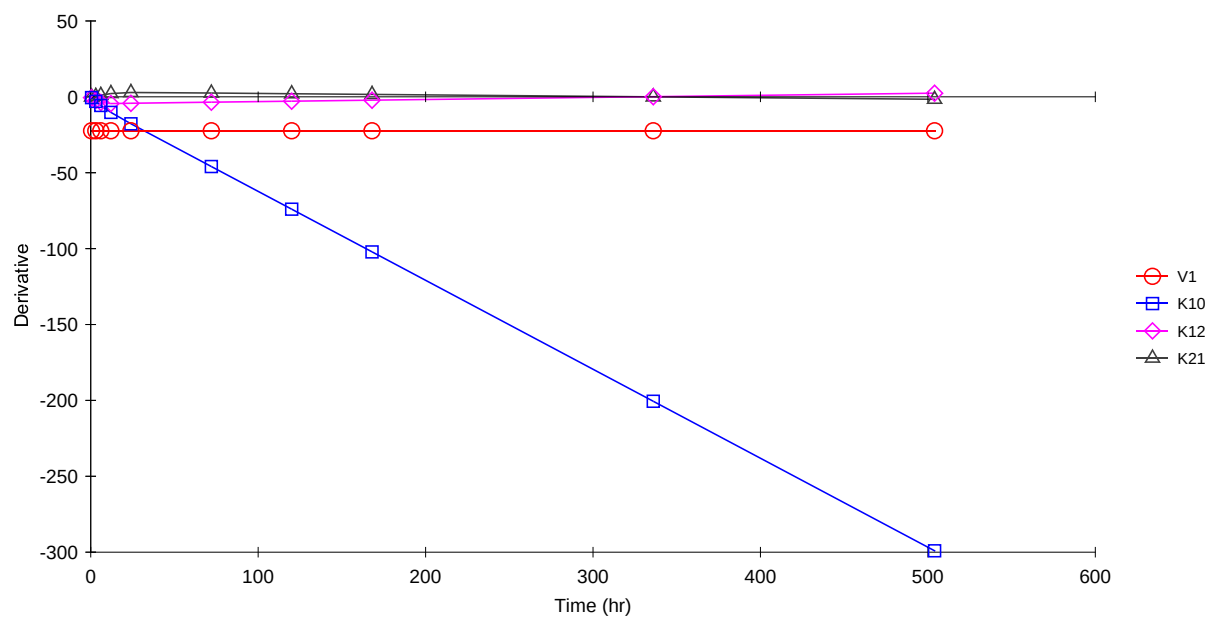
Weighted Predicted Y vs. Weighted Residual Y



X vs. Weighted Residual Y



Partial Derivatives



REPORT NUMBER: LT3114-PHA-015-R	
Title: Pharmacokinetic evaluation of B3 in C57Bl/6 mice	Page 16 of 20

Appendix 2

30mg/kg subcutaneous injection of B3 in C57Bl/6 mice WinNonlin Compartmental Modeling Analysis

WinNonlin compiled model:

NCA Model 200 - Extravascular Input for Plasma Data

Settings for analysis:

Input Workbook: B3 Mouse PK

Input Worksheet: SubQ 30 mg/kg

Input Sort Keys: [none]

Linear trapezoidal (Linear Interpolation) method used

Input data:

Time (hr)	Conc (mg/L)
0.5	2.6
3	112.1
6	380.9
12	265.1
24	417.2
72	398.5
120	219.1
168	178.1
336	127.8
504	52.9

Final Parameters

Output data:

MRTIN	242.7810
MRTIN	241.1512
MRTlas	170.7832
AUMC	38.4636
AUMCI	25455715
AUMC	37.8469
AUMCI	25203156
AUMCI	15664533
Cl F dr	0.0003
Vz F d	0.0692
AUC %	12.5214
AUCIN	3495.017
AUCIN	104850.5
Cl F ob	0.0003
Vz F o	0.0694
AUC %	12.2380
AUCIN	3483.728
AUCIN	104511.8
AUCall	91721.72
AUClast	91721.72
Clast	52.9000
Tlast	504.0000
Cmax	13.9067
Cmax	417.2000
Tmax	24.0000
Tlag	0.0000
HL La	167.5888
Lambda	504.0000
Lambda	24.0000
Lambda	0.0041
No poin	6.0000
Corr X	-0.9745
Rsq adi	0.9371
Rsq	0.9496

No values excluded from lambda_z calculation

Non-Transposed Final Parameters

Parameter	Units	Estimate
Rsqr		0.9496
Rsqr_adjusted		0.9371
Corr_XY		-0.9745
No_points_lambda_z		6.0000
Lambda_z	1/hr	0.0041
Lambda_z_lower	hr	24.0000
Lambda_z_upper	hr	504.0000
HL_Lambda_z	hr	167.5888
Tlag	hr	0.0000
Tmax	hr	24.0000
Cmax	mg/L	417.2000
Cmax_D	kg*mg/L/mg	13.9067
Tlast	hr	504.0000
Clast	mg/L	52.9000
AUClast	hr*mg/L	91721.7250
AUCall	hr*mg/L	91721.7250
AUCINF_obs	hr*mg/L	104511.8635
AUCINF_D_obs	hr*kg*mg/L/mg	3483.7288
AUC_%Extrap_obs	%	12.2380
Vz_F_obs	L/kg	0.0694
CL_F_obs	L/hr/kg	0.0003
AUCINF_pred	hr*mg/L	104850.5146
AUCINF_D_pred	hr*kg*mg/L/mg	3495.0172
AUC_%Extrap_pred	%	12.5214
Vz_F_pred	L/kg	0.0692
CL_F_pred	L/hr/kg	0.0003
AUMClast	hr*hr*mg/L	15664533.0750
AUMCINF_obs	hr*hr*mg/L	25203156.8746
AUMC_%Extrap_obs	%	37.8469
AUMCINF_pred	hr*hr*mg/L	25455715.9142
AUMC_%Extrap_pred	%	38.4636
MRTlast	hr	170.7832
MRTINF_obs	hr	241.1512
MRTINF_pred	hr	242.7810

Dosing

Constant	Value
Dose	30
Time of Last Dose	0

Summary Table

Time (hr)	lambda_z_ Incl	Conc (mg/L)	Predicted (mg/L)	Residual (mg/L)	AUC (hr*mg/L)	AUMC (hr*hr*mg/L)	Weight
0.0000		0.0000			0.0000	0.0000	0.0000
0.5000		2.6000			0.6500	0.3250	0.0000
3.0000		112.1000			144.0250	422.3250	0.0000
6.0000		380.9000			883.5250	4354.8750	0.0000
12.0000		265.1000			2821.5250	20754.6750	0.0000
24.0000	*	417.2000	395.3676	21.8324	6915.3250	99918.6750	1.0000
72.0000	*	398.5000	324.1764	74.3236	26492.1250	1028833.8750	1.0000
120.0000	*	219.1000	265.8042	-46.7042	41314.5250	2348449.8750	1.0000
168.0000	*	178.1000	217.9427	-39.8427	50847.3250	3697557.0750	1.0000
336.0000	*	127.8000	108.7862	19.0138	76542.9250	9817931.4750	1.0000
504.0000	*	52.9000	54.3007	-1.4007	91721.7250	15664533.0000	1.0000

X vs. Observed Y and Predicted Y

